AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

T316
Advanced Networking

Final Examination
Fall Semester 2024/2025

Date: 11/Jan./2025

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Number of Exam Pages: 4 (including this cover sheet)	Time Allowed: 3 hours

Instructions:

- Total Marks: **100**
- This exam consists of **3** parts.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Calculators** are allowed.

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PART 1: Multiple Choices questions (10 x 2 = 20 Marks)

Question: Choose the correct Answer.

1. Which wireless technology typically has the shortest range?
a) Wi-Fi b) Bluetooth c) NFC d) ZigBee
2. What is the maximum transmission distance of a single-mode fiber optic cable?
a) 500 meters b) 2 kilometers c) 10 kilometers d) 100+ kilometers
3. Which of the following is NOT a characteristic of 5G networks?
a) Higher data rates b) Lower latency
c) Increased energy efficiency d) Decreased network capacity
4. Which protocol is used for secure remote login to network devices?
a) HTTP b) FTP c) SSH d) SMTP
5. Which of the following is NOT a benefit of network virtualization?
a) Improved scalability b) Increased hardware dependency
c) Enhanced security d) Reduced operational costs
6. Which of the following is NOT a characteristic of Software-Defined Networking?
a) Centralized network intelligence b) Programmability
c) Hardware-centric approach d) Network abstraction
7. What is the primary purpose of a Network Address Translation (NAT) device?
a) To encrypt network traffic b) To filter incoming packets
c) To conserve IP addresses d) To increase network speed
8. What is the primary function of VoIP technology?
a) To transmit voice over traditional telephone lines
b) To enable video conferencing only
c) To allow voice transmission over Internet Protocol-based networks
d) To improve the quality of analog phone calls
9. In an edge computing model, where does data processing primarily occur?
a) In the cloud b) On a nearby edge device
c) Always in a remote data center d) Only on the IoT device itself
10. What is the key capability of SDN (Software Defined Networking)?
a) It has no effect on network infrastructure management
b) It can only control hardware-based networking
c) It enables programmatic and holistic control of network infrastructure
d) It is exclusively used for cloud computing

PART 2: Short Answer Questions (30 Marks)

Question 1 (10 marks)

A) MIMO technology is fundamental to modern wireless systems. Describe what MIMO means and explain how it differs from traditional single-antenna transmission systems. (4 Marks)

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B) Compare and contrast CDMA (Code Division Multiple Access) with TDMA (Time Division Multiple Access), highlighting the key advantages of each approach as illustrated in Table P2Q1. (6 Marks)

Aspects	CDMA	TDMA
Principle	1 mark	1 mark
Channel usage	1 mark	1 mark
Advantage	1 mark	1 mark

Table P2Q1: CDMA vs TDMA

Question 2 (10 marks)

A) Based on the network topology shown in Figure P2Q2, describe how BGP would be configured and operated to enable full routing connectivity between AS 1, AS 2, and AS 3. (5 marks)

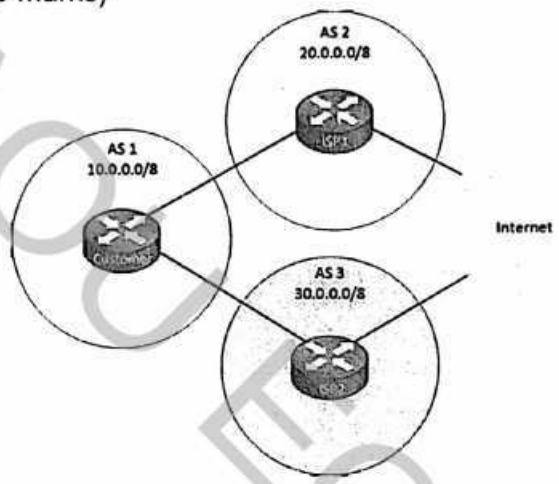


Figure P2Q2: Network Scenario

B) Explain the types of BGP sessions that would be established between the routers. (5 Marks)

Question 3 (10 marks)

A large healthcare organization with multiple hospitals is planning to modernize its network infrastructure using Software-Defined Networking (SDN).

A) Describe two essential SDN components that would be crucial for managing critical healthcare network operations. (5 marks)

B) Explain two ways SDN could improve patient data security and compliance across the hospital network. (5 marks)

PART 3: Long/Problem Solving Questions (50 Marks)

Question 1: (10 marks)

A) Discuss **two** factors that influence indoor wireless signal propagation in modern buildings. (4 marks)

B) Explain how adaptive modulation techniques help overcome wireless communication challenges. (3 marks)

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C) Describe one technique used in channel estimation to improve signal detection. (3 marks)

Question 2: (15 marks)

A) What does it mean for a wireless network to be operating in "infrastructure mode?" If the network is not in infrastructure mode, what mode of operation is it in, and what is the difference between that mode of operation and infrastructure mode? (4 marks)

B) In wireless networks, describe two ways the MAC sublayer manages: i) Channel access ii) Collision avoidance (6 marks)

C) Study the timing diagram depicted in Figure P3Q2 and answer the following questions:

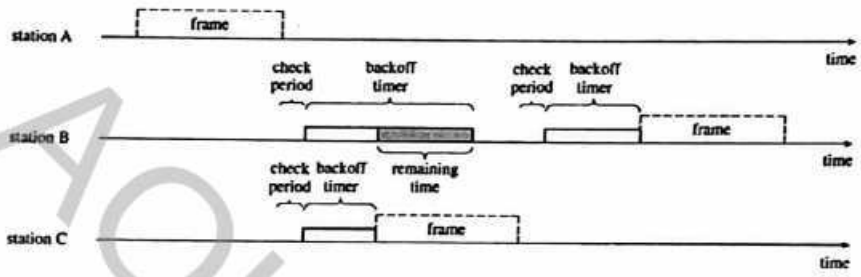


Figure P3Q2: Timing diagram

A) When Station A is transmitting its frame, what are Stations B and C doing? (2.5 marks)

B) Why does Station B need two check periods and backoff timers before transmitting? (2.5 marks)

Question 3: (15 marks)

A) Define Self-Organizing Networks (SON) and explain their key functionalities. (6 marks)

B) Discuss **three** significant challenges in implementing SON features. (6 marks)

C) How does SON improve network operation and maintenance? (3 marks)

End of questions

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